

Louis GEILER, PhD

Quantitative Analyst

Equity Factor Models · Market Impact · Machine Learning · Python / q/kdb+

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Quantitative analyst with a PhD in ML and a double degree in applied mathematics & quantitative finance (Dauphine / CY Tech). Built and deployed production-grade equity factor models, market impact platforms and P&L analytics at Morgan Stanley. Strong background in statistical modelling, time-series analysis and applied ML research.

EXPERIENCE

Quantitative Analyst — RiskLab (Equities, IED)

Morgan Stanley

Jan 2023 – Present (3 yr 3 mo)

Paris, France

FACTOR MODELS & RESEARCH

- Re-implemented the **MSCI Barra equity factor model** for the global book — factor-return estimation via constrained weighted linear regression and other optimisation approaches, Delta P&L decomposition by style / industry / country / region with drill-downs and MAD-based reconstruction error monitoring.
- Designed a **scenario analytics tool** to stress-test equity portfolios under spot price shocks ($\pm 10\%$, $\pm 50\%$ grid) — computed Barra meta-factor deformations under each scenario to measure factor sensitivity changes across aggregation levels.
- Built a **factor exposure monitoring system** for the global equity universe — computed Delta Cash $\times$ factor exposure per stock subset, aggregated across time; optimised computational architecture (pre-computation, parallelisation) to handle large-scale data volumes.

RISK ANALYTICS & PRODUCTION SYSTEMS

- Built a production-grade **liquidity & market impact platform** (1.5 yr) — data pipeline generating consolidated volume, MDV21, prices, volatility and a full suite of liquidity decomposition metrics across global equity markets (US, Europe, Asia); led an exploratory research study on **volume/volatility regime dynamics** for stress-testing; benchmarked and deployed 6 internal market impact models; implemented **cross-impact methods**.
- Designed a 3-step **Greek P&L monitoring framework** — (i) trend/regime detection, (ii) multi-trader benchmarking, (iii) anomaly detection: univariate rolling MAD on Delta/Vega series + multivariate **Isolation Forest** with feature engineering, replacing a manual process taking several hrs/week.
- Designed a **Delta Cash attribution framework** — decomposed daily P&L changes via a Taylor expansion into market-driven (Delta, Gamma), option-expiry and position-change effects; enriched with intraday trade count and average cost per trade for multi-trader profiling.

Factor models

Liquidity analytics

Anomaly detection

Time series

Optimisation

Python (pandas, polars, FastAPI, Luigi)

q/kdb+

Git

Parallel computing

Jenkins (CI/CD)

Data Scientist / PhD CIFRE

Brigad

Jan 2019 – Sep 2022 (3 yr 9 mo)

Paris, France

RESEARCH & PUBLICATIONS

- Led research on **tabular ML for churn prediction**: comparative study of tree-based models, neural nets and semi-/self-supervised representation learning.
- Published **3 peer-reviewed papers + 1 preprint** (JDSA, EGC, DKE) on churn prediction, customer profiling and contrastive learning for tabular data.

DATA SCIENCE & ENGINEERING

- Built end-to-end **churn and engagement modelling pipelines**: SQL extraction, feature engineering, evaluation and delivery of actionable scores to stakeholders.
- Designed **segmentation / persona classification** combining supervised methods with RFM-style business segmentation, exposed via FastAPI.
- Developed **demand/GMV forecasting** prototypes under strong regime shifts (COVID) using time-series models.

Tabular ML

Classification

Semi-supervised learning

Time series

Python (pandas, FastAPI)

SQL

Data Scientist / VIE

Schneider Electric

Oct 2017 – Nov 2018 (1 yr 1 mo)

Kuala Lumpur, Malaysia

- Autonomous data science role across SE Asia manufacturing sites: **predictive maintenance**, computer vision quality inspection, supply-chain analytics (demand forecasting, supplier risk) and IIoT use cases.

Predictive maintenance

Classification

Computer vision

Time series

Python

Tableau

Data Scientist · Swiss Life, Levallois-Perret · Mar – Aug 2017 (6 mo)

- POC data science in insurance marketing — churn modelling, propensity scoring, fraud detection (NLP/ document parsing) and topic discovery on NPS verbatims.

Data Scientist / Intern · Société Générale, La Défense · May – Oct 2016 (6 mo)

- NLP/text analytics — email classification (word embeddings, SVM), topic modelling and legal document parsing (NER, OCR).

EDUCATION

PhD CIFRE — Machine Learning

2019 – 2022

Université Paris Cité

Tabular ML for churn prediction; tree-based models, neural networks, representation learning, semi-/self-supervised learning. 3 publications + 1 preprint (JDSA, EGC, DKE).

Teaching & supervision: mentored 4+ student groups on applied ML projects (sentiment analysis).

Engineer's Degree — Applied Mathematics & Data Science **Honors**

2013 – 2016

CY Tech (ex-EISTI) — double degree with Université Paris Dauphine

ML, statistics, numerical analysis, time series, NLP, quantitative finance, portfolio management.

Master — Financial Informatics (MIAGE-IF) **Honors**

2016

Université Paris Dauphine

Portfolio management, quantitative finance, optimisation for finance, Big Data (Hadoop/Spark).

Master 1 — Operations Research (Recherche Opérationnelle)

2015

Université Paris Dauphine

Combinatorial optimisation, advanced linear programming, symbolic AI. Projects: TSP & portfolio allocation in C++ with CPLEX.

SKILLS & TOOLS

Quantitative Finance

Asset pricing · Equity factor models (Barra-style) · P&L decomposition · Liquidity analytics / Market impact / Cross-impact · Scenario / stress-testing

Machine Learning

Tabular ML · Tree-based models · Neural networks (Generative, NLP, Vision) · Self-supervised & representation learning · Clustering · Probabilistic graphical models · Interpretability (SHAP, LIME)

Statistics & Optimisation

GLM · Time series · Anomaly detection · Conformal prediction · Linear, combinatorial & quadratic / non-linear optimisation

Programming

Python (pandas, polars, scikit-learn, PyTorch) · q/kdb+ · SQL

Data & Engineering

Luigi · Parquet · GPFS · Autosys · Parallel computing · FastAPI

Software Practices

Git · Jenkins (CI/CD) · Unit testing · Production monitoring · Logging & debugging

Languages

French (native) · English (fluent)

INTERESTS

Sport: CrossFit, Running · Dance: Salsa